

Disproof of Conjecture 00000007612

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Claim under test. The conjecture asserts, without a parity restriction, that the dimension of an irreducible Clifford-algebra representation is exactly $2^{n/2}$.

Counterexample. Take the complex Clifford algebra on one generator with positive square:

$$\text{Cl}_1(\mathbb{C}) = \mathbb{C}[e]/(e^2 - 1).$$

The map

$$\rho(a + be) = a + b$$

is a unital algebra homomorphism to $\mathbb{C}$, because $\rho(e)^2 = 1$. Thus $\mathbb{C}$ is a nonzero one-dimensional $\text{Cl}_1(\mathbb{C})$ -module. A one-dimensional nonzero module has no proper nonzero subspace, so it is irreducible.

For $n = 1$, however, the proposed formula is

$$2^{n/2} = 2^{1/2} = \sqrt{2} \neq 1.$$

Indeed no natural-number dimension d can satisfy $d^2 = 2$. Hence the claimed dimension law is false. (The familiar correct complex formula is $2^{\lfloor n/2 \rfloor}$ for an irreducible module, with the usual odd/even qualification on the algebra.)

Formal verification. The Lean file encodes $a + be$ as an integer pair, its multiplication modulo $e^2 = 1$, and the character $a + be \mapsto a + b$. It proves multiplicativity, nontriviality, and the decisive numerical mismatch $1^2 \neq 2$ for the explicit one-dimensional representation. This is the exact finite algebraic core of the counterexample.